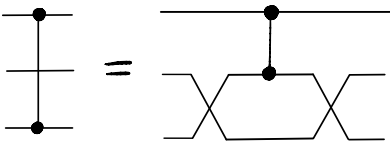
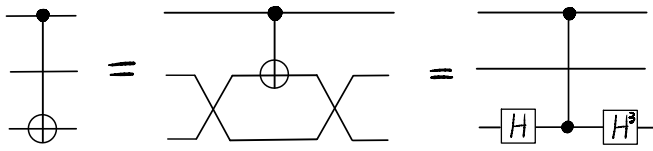
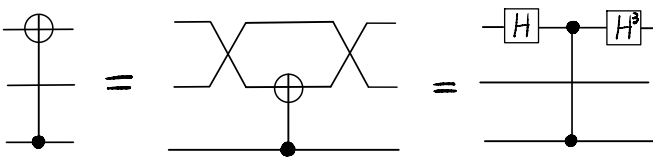


Three - Qudit Derived Relations

D7: $rCZ :=$ 

D8: $rCX :=$ 

D9: $rXC :=$ 

C_1 :

$=$

C'_{17} : 

C_{17}^2 :

C_{17}^3 : 

C_{17}^4 : 

C_{17}^5 : 

C_{17}^6 :

C_{17}^7 :

Diagrammatic equation for C_{17}^7 :

$$\text{Diagram 1} = \text{Diagram 2}$$

C_{17}^8 :

C_{17}^9 :

$C_{17}^{10} :$

C_{17}'' :

The diagram shows a 4-qubit circuit. The top two qubits are the control lines for two Toffoli gates. The bottom two qubits are the target lines for two CNOT gates. The first CNOT gate has its control on the top qubit and target on the bottom qubit. The second CNOT gate has its control on the second qubit and target on the third qubit. The first Toffoli gate has controls on the top two qubits and target on the third qubit. The second Toffoli gate has controls on the top two qubits and target on the fourth qubit.

C_{17}^{12} :

C_{17}^{13} :

C_{17}^{14} :

Diagrammatic equation for C_{17}^{14} showing a crossing with a dot and a circle on the left, and the same crossing with the dot and circle swapped on the right, separated by an equals sign.

C_{17}^{15} :

The diagram shows two equivalent circuit representations of the commutator C_{17}^{15} . On the left, a control dot is on the bottom line and a target circle is on the top line. On the right, the control dot is on the top line and the target circle is on the bottom line. The two diagrams are separated by an equals sign, indicating they are equivalent.

C_{17}^{16} :



C_{17}^{17} :

C_{17}^{18} :

$C_{17}^{24} :$

C_{17}^{19} :

C_{17}^{25} :

C_{17}^{20} :

Diagrammatic equation for C_{17}^{20} :

Left side: A horizontal line with a dot is connected by a vertical line to a circle on another horizontal line.

Right side: A horizontal line with a dot is connected by a vertical line to a circle on another horizontal line.

C_{17}^{26} :

Diagrammatic equation for C_{17}^{26} :

$$\begin{array}{c} \bullet \\ | \\ \bigoplus \end{array} \text{ crossing} = \text{ crossing} \begin{array}{c} | \\ \bullet \\ \bigoplus \end{array}$$

C_{17}^{21} :

Diagrammatic equation for C_{17}^{21} :

$$\text{Diagram 1} = \text{Diagram 2}$$

Diagram 1: A vertical line with a dot on the top wire and a circle on the bottom wire, connected to a crossing.

Diagram 2: A crossing followed by a vertical line with a dot on the top wire and a circle on the bottom wire.

C_{17}^{27} :



C_{17}^{22} :

C_{17}^{28} :

C_{17}^{23} :

$$\text{Diagram 1} = \text{Diagram 2}$$

C_{17}^{29} :

$$C_{18}: \begin{array}{c} \bullet \\ \oplus \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \oplus \\ \bullet \end{array} \quad \text{Lem 58} \quad \checkmark \checkmark \checkmark$$

$$C_{18}^1: \begin{array}{c} a \\ \oplus \\ \bullet \end{array} = \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \quad \text{Lem 59} \quad \checkmark \checkmark \checkmark$$

$$C_{18}^2: \begin{array}{c} \bullet \\ \oplus \\ \bullet \end{array} \begin{array}{c} a \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \oplus \\ \bullet \end{array} \begin{array}{c} a \\ \bullet \end{array} \quad \text{Lem 60} \quad \checkmark \checkmark \checkmark$$

$$C_{18}^3: \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \begin{array}{c} b \\ \bullet \end{array} = \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \begin{array}{c} b \\ \bullet \end{array} \quad \text{Lem 61} \quad \checkmark \checkmark \checkmark$$

$$C_{18}^4: \begin{array}{c} \bullet \\ \oplus \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \oplus \\ \bullet \end{array} \quad rCZ^{-1} \quad \checkmark \checkmark \checkmark$$

$$C_{18}^5: \begin{array}{c} a \\ \oplus \\ \bullet \end{array} = \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \quad rCZ^{-a} \quad \checkmark \checkmark \checkmark$$

$$C_{18}^6: \begin{array}{c} a \\ \oplus \\ \bullet \end{array} = \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \quad rCZ^{-a} \quad \checkmark \checkmark \checkmark$$

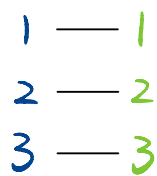
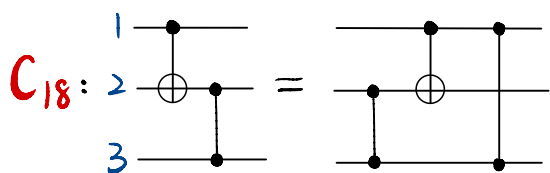
$$C_{18}^7: \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \begin{array}{c} b \\ \bullet \end{array} = \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \begin{array}{c} b \\ \bullet \end{array} \quad rCZ^{-ab} \quad \checkmark \checkmark \checkmark$$

Corollary 62

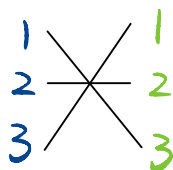
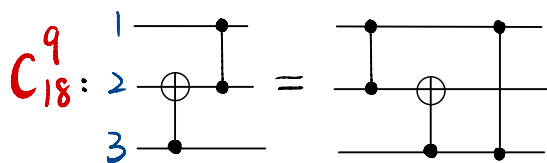
$$C_{18}^{8-12}: \begin{array}{c} \bullet \\ \oplus \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \oplus \\ \bullet \end{array} \quad CX^{-1} \quad CZ^{-1} \quad \checkmark \checkmark \checkmark \quad \text{Lem 63} \quad \checkmark \checkmark \checkmark$$

$$C_{18}^{8-3/4}: \begin{array}{c} a \\ \oplus \\ \bullet \end{array} = \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \quad CX^{-a} \quad CZ^{-1} \quad CX^a \quad \checkmark \checkmark \checkmark = \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \quad CX^a \quad CZ^{-1} \quad CX^{-a} \quad \checkmark \checkmark \checkmark \quad \text{Lem 64} \quad \checkmark \checkmark \checkmark$$

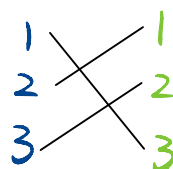
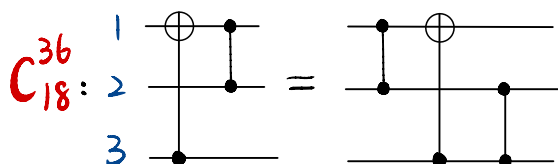
$$C_{18}^{8-5/6}: \begin{array}{c} a \\ \oplus \\ \bullet \end{array} = \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \quad CX^{-1} \quad CZ^{-a} \quad \oplus \quad CZ^a \quad \checkmark \checkmark \checkmark = \begin{array}{c} a \\ \oplus \\ \bullet \end{array} \quad CZ^a \quad CX^{-1} \quad CZ^{-a} \quad \checkmark \checkmark \checkmark \quad \text{Lem 65} \quad \checkmark \checkmark \checkmark$$



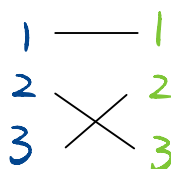
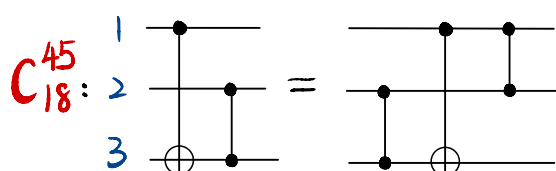
$(1, 2, 3)$



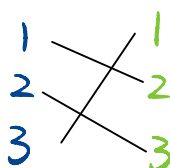
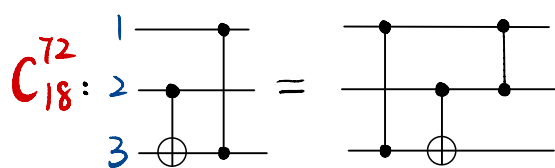
$(3, 2, 1)$



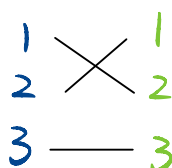
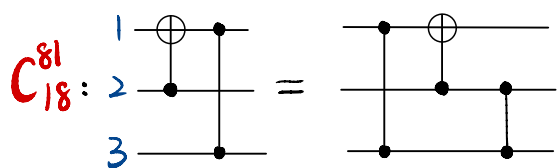
$(2, 3, 1)$



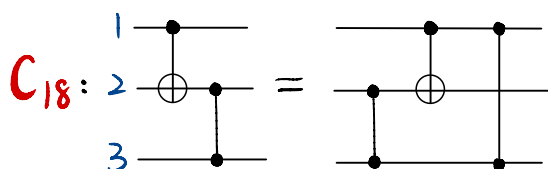
$(1, 3, 2)$



$(3, 1, 2)$



$(2, 1, 3)$



$\Rightarrow$

